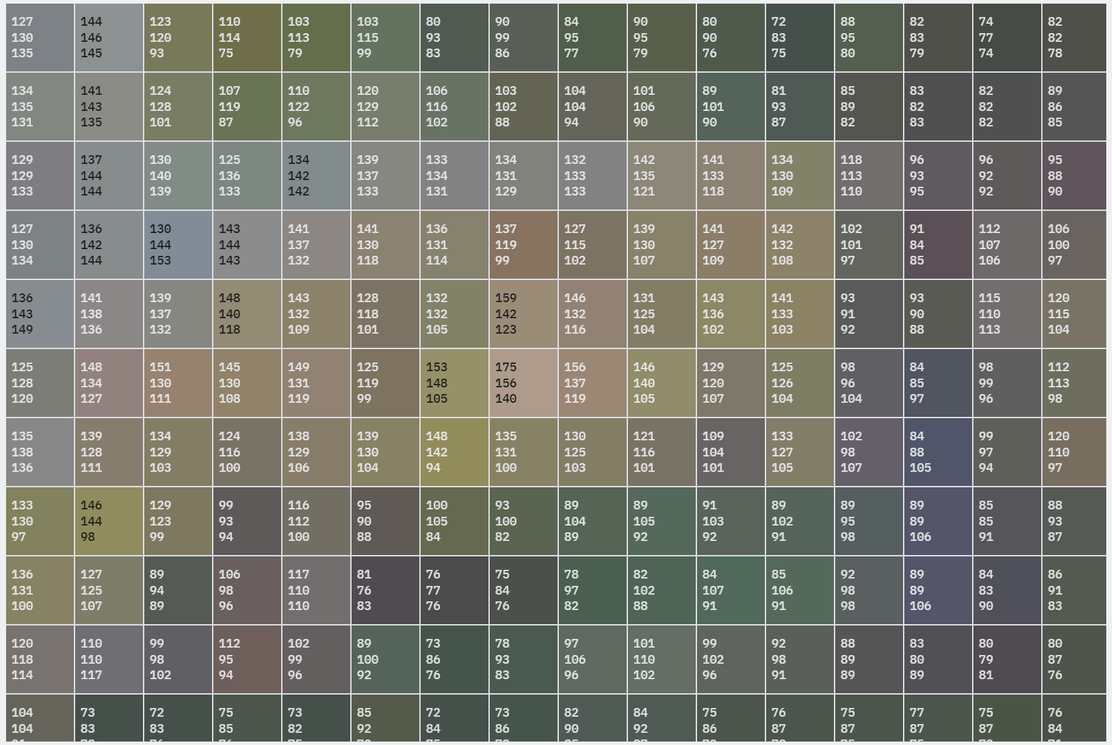
Art Data & Data Art

Reading art with data · Making art from data

Interactive presentation: caglarfurkan.github.io/va430-data-and-art ↗



Georges Seurat, *A Sunday on La Grande Jatte — 1884*, 1884–86. Art Institute of Chicago, 1926.224. Public domain.

20 October 2026 · SUSAM G010

Furkan Çağlar

Four parts, one hour.

13 min

Data, visualization, tools

What data is, how charts work, who makes data art, and with what

15 min

Art data

Questions, open museum tables, cleaning messy dates, images as numbers

22 min

Live painting exercise

Rate one painting on your phone, hear its context, rate it again, and build a live concept network

10 min

Examples & wrap-up

Works from my own practice, four ways to work with data, and a starter kit

Data is a record of an observation.

Someone looked, decided what mattered, and wrote it down.

Georges Seurat

French, 1859–1891

A Sunday on La Grande Jatte — 1884

1884–86, border added 1888–89

Oil on canvas

207.5 × 308.1 cm

Helen Birch Bartlett Memorial Collection, 1926.224

artist	born–died	title	date	medium	h × w (cm)	id
Georges Seurat	1859–1891	A Sunday on La Grande Jatte — 1884	1884–86	Oil on canvas	207.5 × 308.1	1926.224

What the table leaves out

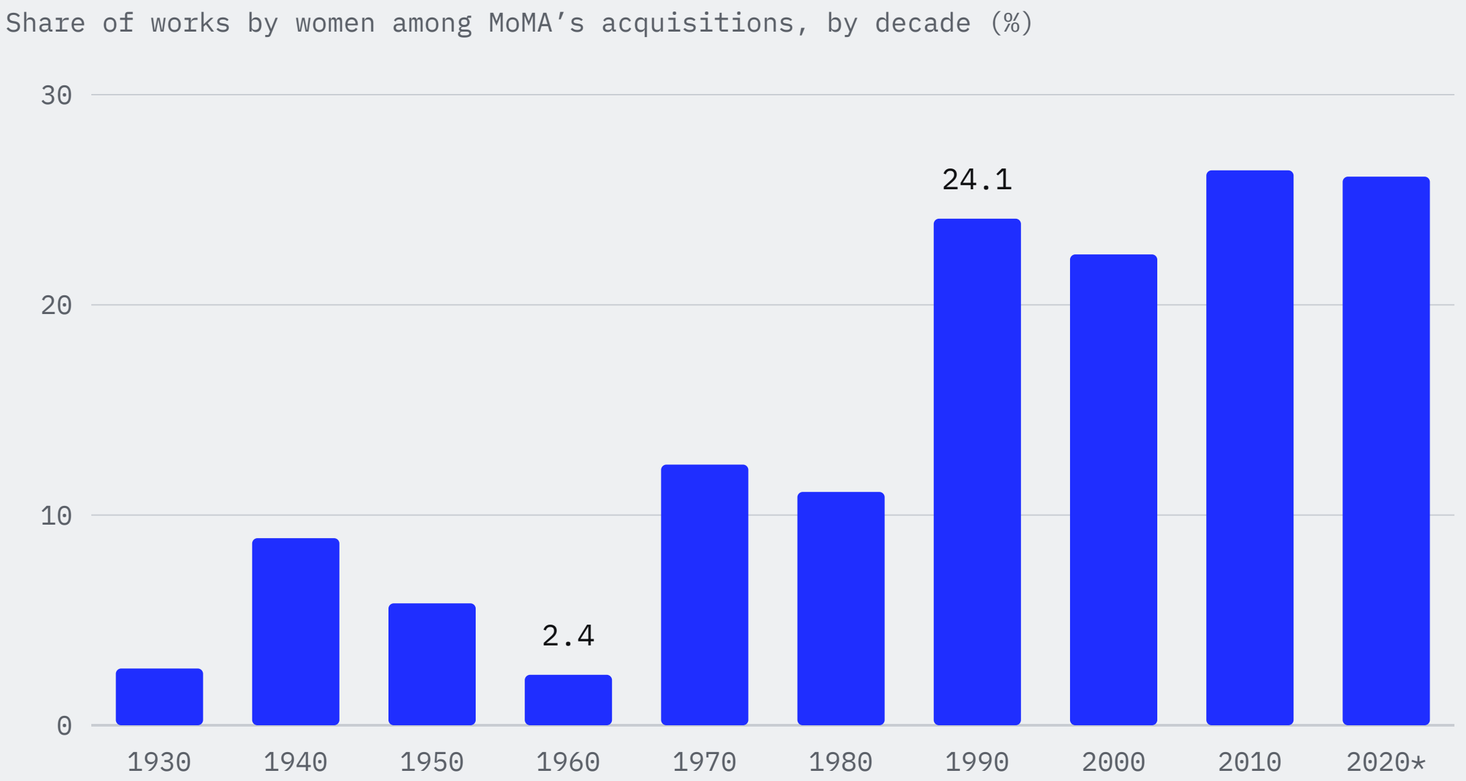
- Standing in front of it in the room (it is over three metres wide)
- Two years of placing tiny dots by hand, plus the dotted frame Seurat painted later
- Who the people on the island are, and why nobody is talking to anyone

Label: Art Institute of Chicago collection record, via its public API (1926.224).

A visualization maps data to marks and channels.

Length	bars
Position	dots
Slope	line
Area	circles
Color	strip

Length from a common baseline. Easy to compare any two decades.



Data: MoMA Collection (GitHub, CC0), 160,700 records; first listed artist with a recorded gender. 2020s = 2020–2026.

Six forms, all older than computers.



Bars

compare amounts

W. E. B. Du Bois and students, Paris, 1900



Timeline

show lives and periods

Joseph Priestley, A Chart of Biography, 1765



Polar area

parts of a whole over time

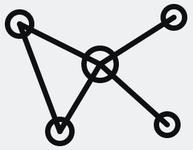
Florence Nightingale, 1858



Flow

movement and loss

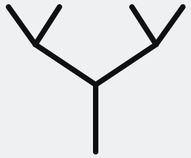
Charles Joseph Minard, 1869



Network

who is connected to whom

Alfred H. Barr Jr., 1936; Mark Lombardi, 1990s



Tree

order knowledge into branches

Diderot and d’Alembert, Encyclopédie, 1751

When the chart becomes the work.

Federica Fragapane

The Stories Behind a Line, 2018

The routes of six asylum seekers to Italy, drawn day by day with their own stories.

Giorgia Lupi & Stefanie Posavec

Dear Data, 2014–15

A year of weekly hand-drawn postcards about everyday habits, sent between New York and London.

dear-data.com ↗

Nadieh Bremer & Shirley Wu

Data Sketches, from 2016

Twelve custom web visualizations, published together with all the messy sketches and code behind them.

datasketch.es ↗

Lev Manovich, Moritz Stefaner et al.

Selfiecity, 2014

3,200 Instagram selfies from five cities, sorted by head tilt, mood, and demographics.

selfiecity.net ↗

Aaron Koblin

The Sheep Market, 2006

10,000 sheep drawn by Mechanical Turk workers, each paid $0.02 to “draw a sheep facing to the left.”

thesheepmarket.com ↗

Refik Anadol

Unsupervised, MoMA, 2022–23

A machine learning model trained on MoMA’s public collection archive, shown in the museum lobby.

refikanadol.com ↗

Nathalie Miebach

Weather sculptures

Storm and weather readings translated into woven reed baskets and musical scores.

nathaliemiebach.com ↗

Burak Arıkan

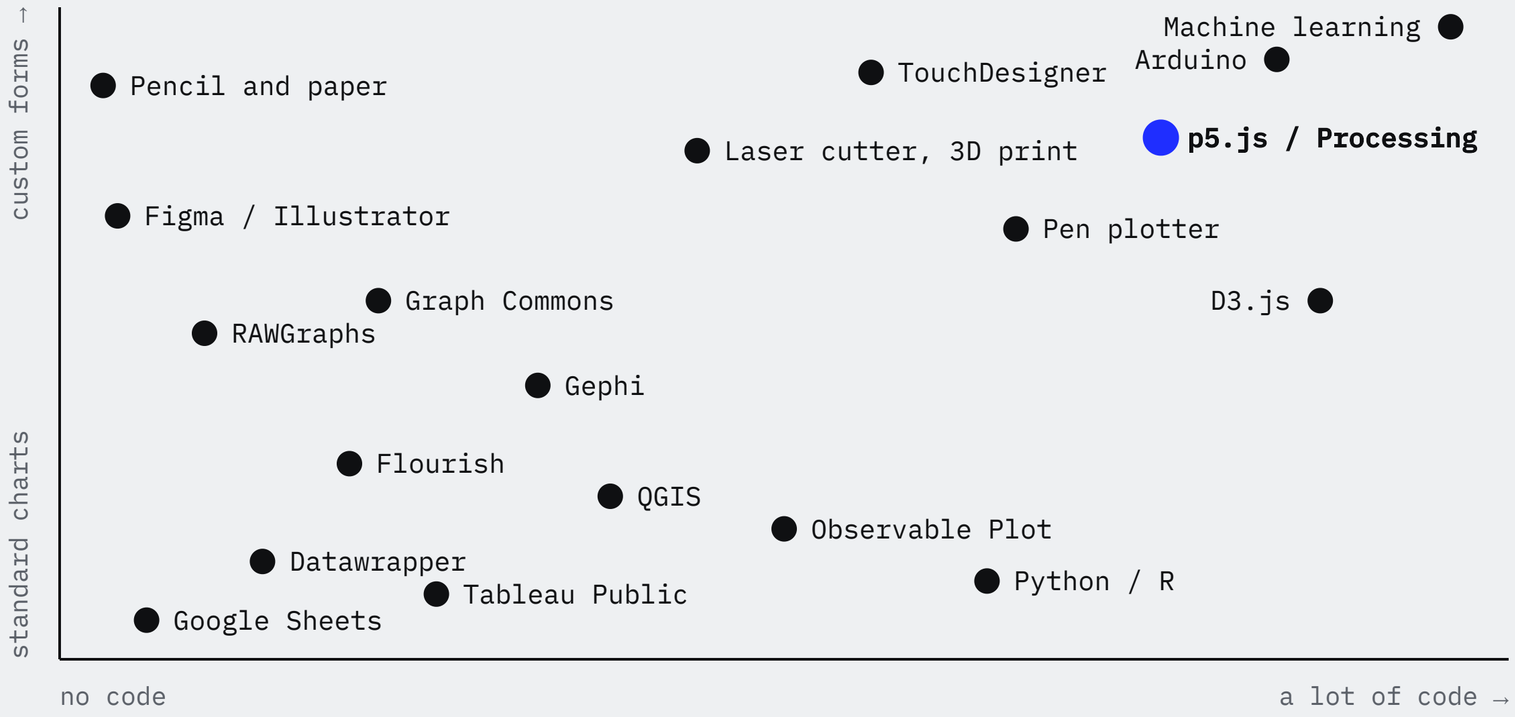
Networks of Dispossession, 2013

Maps of capital, urban projects, and state contracts in Turkey, built collectively on Graph Commons.

graphcommons.com ↗

What do people make it with?

I want to make a **anything** | print | screen | object | drawing



p5.js / Processing

Creative coding: drawing with data, generative form. The next slide runs p5.js.

screen | print | object

> Aaron Koblin: The Sheep Market was made with Processing.

Horizontal axis: how much code you write. Vertical axis: how freely you can shape the visual form. Click any dot.

From chart to artwork.

Edit any number or color below — updates live

Reset code

```
// <-- CHANGE THESE VALUES LIVE ON STAGE:
const scale       = 1.00;      // <-- overall size (try 0.7 to 1.4
const pctPerRing  = 4.5;       // <-- Isolines: % women per blue r
const ringSteps   = 6;         // <-- Isolines: total contour ring
const barWidth    = 11;        // <-- Spectrum: dept bar width in
const radarSides  = 5;         // <-- Manifold: 5 MoMA departments
const showNumbers = true;      // <-- show exact % and counts (tru

const COL_WOMEN   = "#1F2EFF"; // <-- blue: works by women
const COL_TOTAL   = "#9DA3AC"; // <-- grey: total acquisitions
const COL_SPIKE   = "#E8541E"; // <-- orange: 1960s Mies Archive s

// Return config to the live p5.js renderer:
return { scale, pctPerRing, ringSteps, barWidth, radarSides,
         showNumbers, COL_WOMEN, COL_TOTAL, COL_SPIKE };
```

Bar Chart | 1 · Isolines | 2 · Spectrum | 3 · Manifold

distortion



1 · Isolines (İzohips): Outer ring size = total works acquired in that decade. Each blue contour ring = 5% share by women. Hover any decade for exact counts.

Data: MoMA Collection (GitHub, CC0), 1930s–2020s. The 1960s total (44,796 works) jumps because the Mies van der Rohe Archive arrived at once.

Start here

Start with a question you cannot answer yet.

Do viewers agree on whether a painting is calm or tense?	Today’s exercise · data: a survey of this room
Has MoMA collected more work by women since 1989?	data: MoMA collection, GitHub
Where do Istanbul Biennial artists come from, and has that changed?	data: İKSV biennial archive, by hand
Which museums are visible in the open data that trains AI?	data: Wikidata
How do different museums label the same artist?	data: wall labels you photograph

Art data is more than metadata.

Metadata

who made it, when, medium, dimensions, when the museum bought it

Pixels

color palette, brightness, and edges measured from a photo of the work

Text

wall labels, exhibition catalogues, press reviews, artist interviews



Georges Seurat, *A Sunday on La Grande Jatte — 1884*

People & networks

artists, curators, galleries, collectors, and who exhibits with whom

Market & provenance

who owned the work, auction prices, public or private funding

Audience

how long visitors look, where they stand, or our own classroom survey

It can become: a chart | a map | a network | an image plot | an object | a drawing | a sound

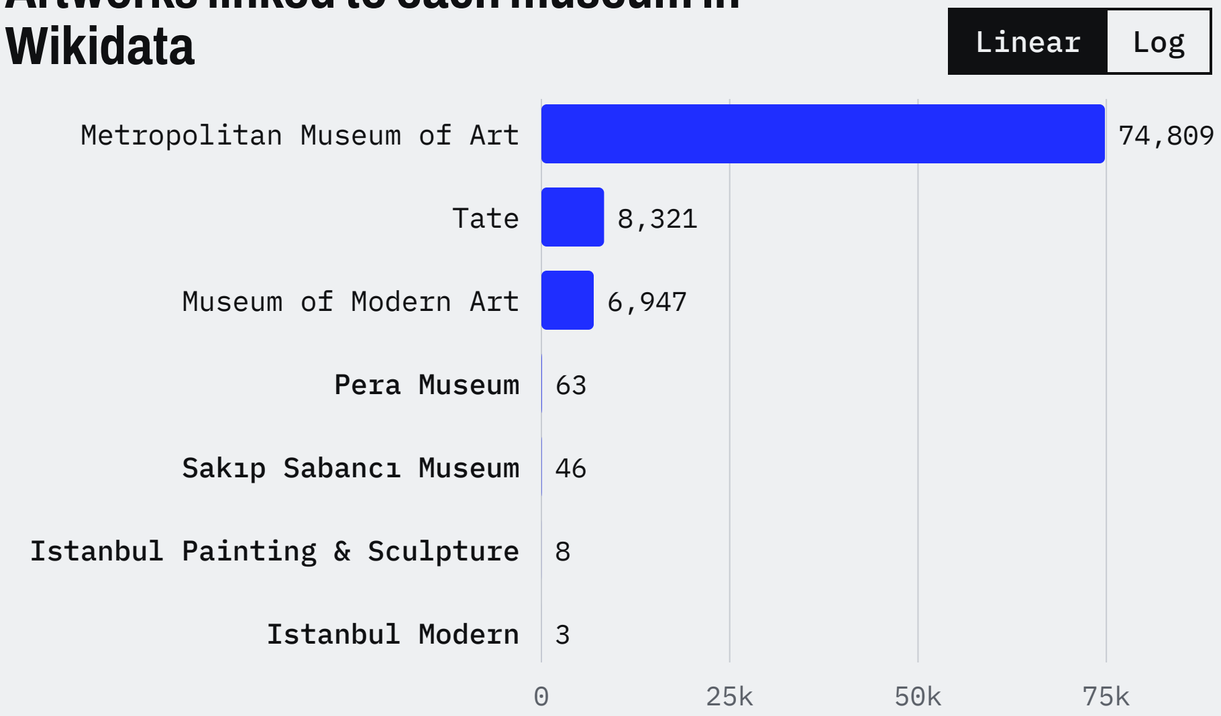
Where to find art data.

Source	What you get	Watch out
MoMA collection `github.com/MuseumofModernArt`	160,700 works as CSV: artist, nationality, gender, dates, acquisition year	Marked “not curatorially approved.” One single archive makes up 10% of all rows.
Met Open Access `CSV on GitHub + Collection API`	480,000+ objects with culture, period, medium, and public-domain images	“Culture” and “period” mix geographic, ethnic, and dynastic labels.
Art Institute of Chicago `api.artic.edu`	133,000 works with styles, dominant colors, and IIIF image links	Style labels are inconsistent (“19th century” vs. “nineteenth century”).
Rijksmuseum `data.rijksmuseum.nl`	High-resolution images and detailed iconographic tags	Check current API access rules and image licenses before batch downloading.
Wikidata `query.wikidata.org`	Artists, works, places, and movements linked together; exports as CSV	Written by volunteers; big Western museums have tens of thousands of rows, local ones have dozens.
Your own survey `forms, field notes, counting`	Built directly around the question you want to ask	Smaller sample size, shaped by who answers—do not collect personal names you do not need.

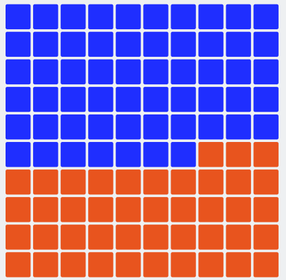
Every collection is a record of what someone chose to keep.

Who is in the data decides what the chart can say.

Artworks linked to each museum in Wikidata



Tate’s collection records



57%

by one artist: J. M. W. Turner. In 1856, 37,711 of his works arrived at once.

Wikidata query, 26 September 2026. Tate Collection data (CC0), 69,201 records, 39,389 by Turner.

Metadata is an interpretation.

Real “Date” values from 3,000 MoMA records → a rule that keeps the first 4-digit year

(c. 1915)	→	**1915**	circa dropped
early 1930s	→	**1930**	a decade, not a single year
1884-86	→	**1884**	range, hyphen
1962–66	→	**1962**	range, en dash
1953—1956	→	**1953**	range, em dash
1971 (printed 2014)	→	**1971**	negative vs. print date
Published 1926 (frottage 1925)	→	**1926**	rule picks publication year
Before 1899	→	**1899**	only an upper limit
n.d.	→	**—**	no date, dropped

633
of 3,000 dates are not a plain year

3
different dash characters

55
listed as “Unidentified” or “Unknown”

Keep the raw column.

Never overwrite what the museum wrote.

Write down your rule.

“First year in the text” is a choice you made; explain it.

Count what you drop.

The rows you leave out are part of the result.

Same image, three methods, different numbers.



Seurat, A Sunday on La Grande Jatte Use another image

or drop one on this slide

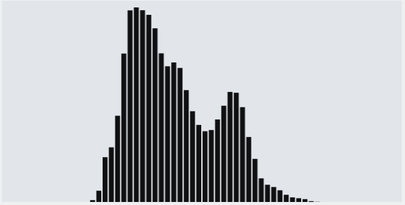
Color

Average | Palette | Hues

#6F7066

the average of every pixel

Light



43 contrast 22

mean brightness, 0 black · 100 white

Structure



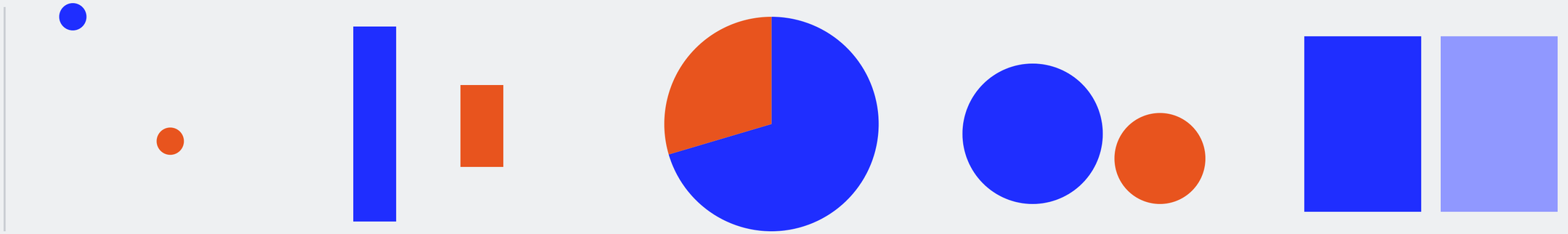
edge threshold 120

10%

of pixels count as an edge

A chart is a claim.

Guess: what percent of A (blue) is B (orange)? The same two values, five ways.



Position

rank 1 · common scale

Length

rank 3 · length, angle

Angle

rank 3 · length, angle

Area

rank 4 · area

Color

rank 6 · saturation

B = 42% of A

New pair

Ranking from William S. Cleveland and Robert McGill, “Graphical Perception,” Journal of the American Statistical Association (1984).

Rate one painting.



Georges Seurat, A Sunday on La Grande Jatte — 1884 (or drop any painting)

or open

caglarfurkan.github.io/va430-data-and-art/p

No names. Ninety seconds.

warm — co

calm — ten

ordered — chaot

flat — de

The gap is not an error. It is the discussion.

raw rows, as they arrive 48

```
time     round,id,cool,tense,chaos,deep
14:44:43 2,stu24,53,68,42,68
14:44:43 1,stu24,47,34,28,66
14:44:43 2,stu23,39,56,27,77
14:44:43 1,stu23,34,20,16,77
14:44:43 2,stu22,47,65,37,72
14:44:43 1,stu22,40,27,23,71
```



Now, the context.



Georges Seurat, A Sunday on La Grande Jatte — 1884

who made it, and how

Two years of studio work after 60+ sketches; Seurat placed pure color dots side by side to mix in the viewer’s eye, then added a painted dot border in 1888–89.

when, where, for whom

An island on the Seine outside Paris: bourgeois Sunday strollers share the bank with working-class rowers; the pet monkey on a leash was a well-known Parisian satire trope.

posture & stillness

Almost every figure stands in strict profile or frontal pose like an ancient frieze—crowded together, yet nobody speaks to or looks at anyone else.

what happened to the paint

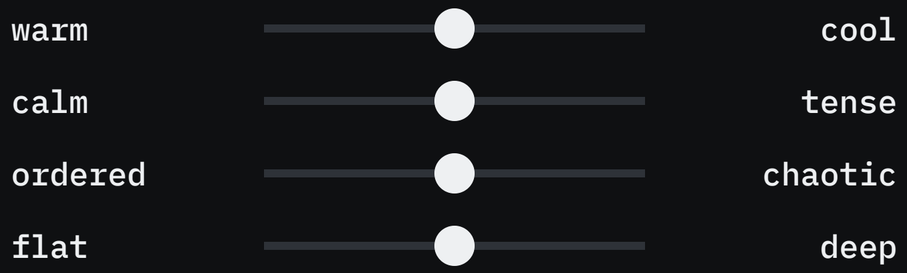
Seurat’s unstable zinc yellow dots darkened to brown within a few years, shifting the warmth and light balance we see today.

Rate it again.

same page on your phone

caglarfurkan.github.io/va430-data-and-art/p

Same four sliders. Has anything moved?

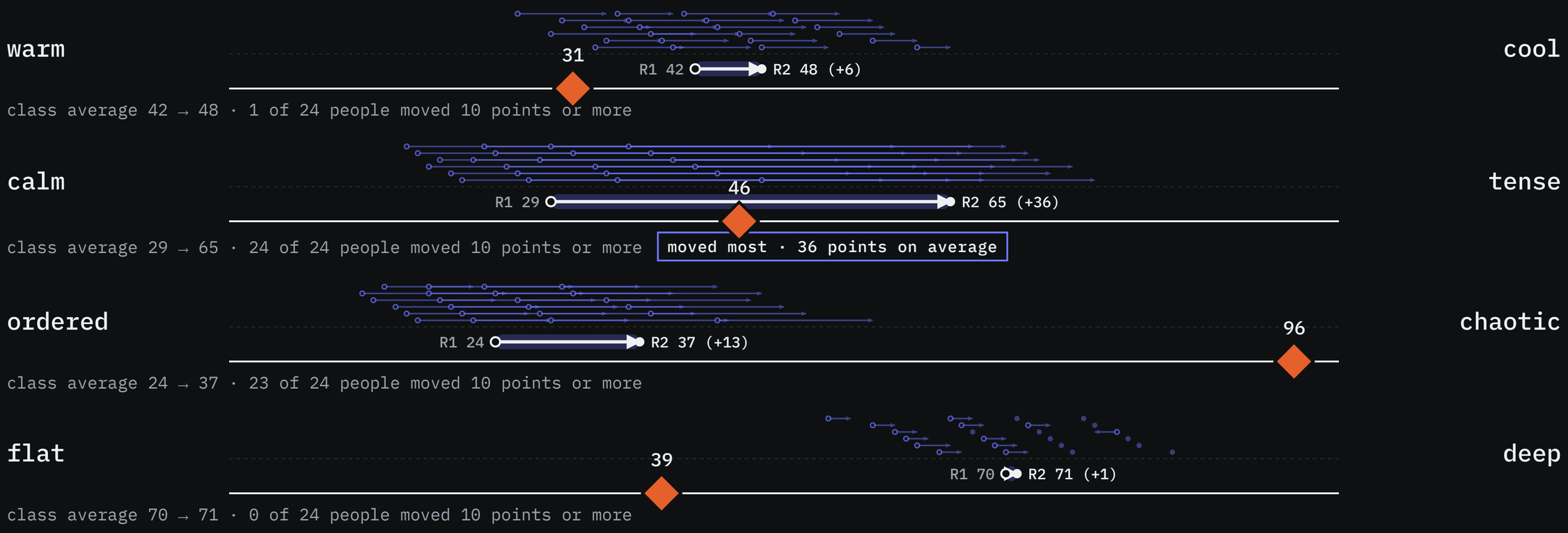


Did context change what we saw?

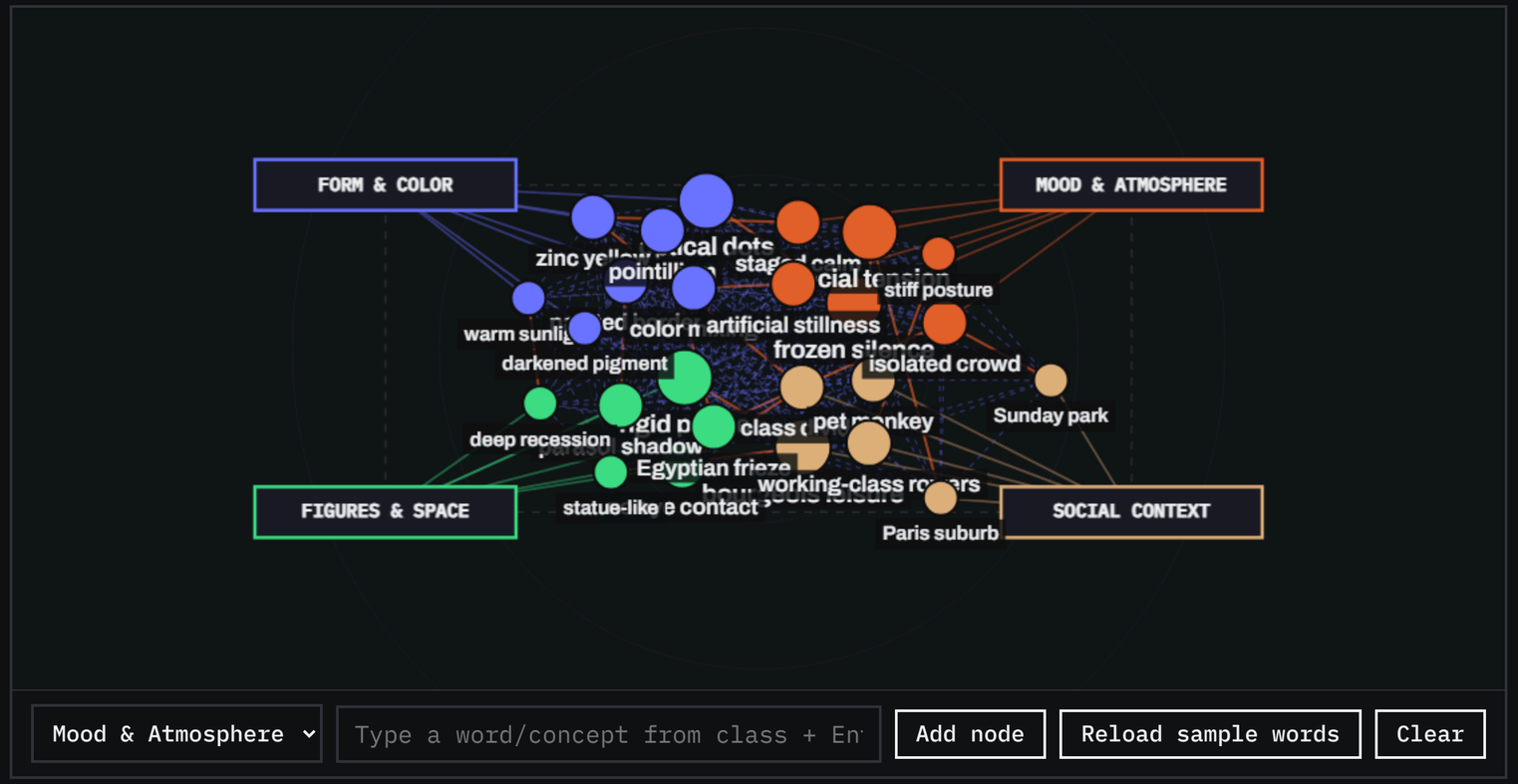
raw rows, as they arrive 48

```
time     round,id,cool,tense,chaos,deep
14:44:43 2,stu24,53,68,42,68
14:44:43 1,stu24,47,34,28,66
14:44:43 2,stu23,39,56,27,77
14:44:43 1,stu23,34,20,16,77
14:44:43 2,stu22,47,65,37,72
14:44:43 1,stu22,40,27,23,71
```

○ Student R1 → R2 — Class average shift ◆ Pixel formula, unchanged



What no slider caught: live concept network.



How the network connects

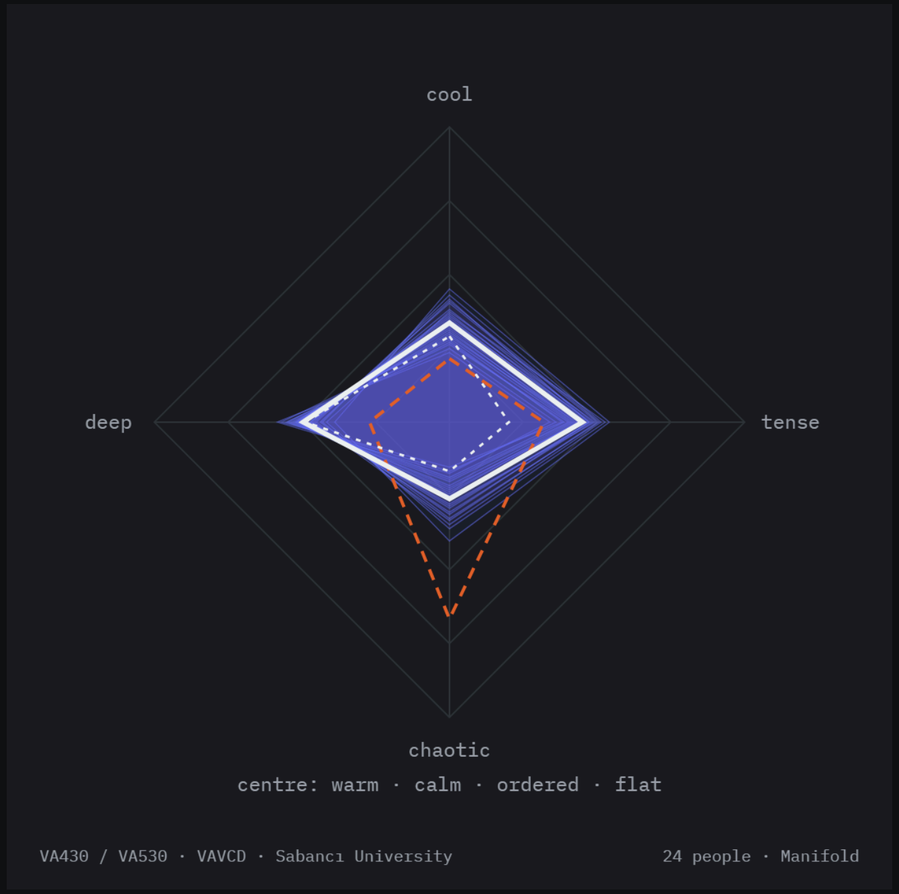
Phones send words in real time. Each word links to its category (Form, Mood, Figures, Context), to other words from the same person, and to classmates who rated the sliders similarly. **Drag any node, or click two nodes to connect them.**

Words from the room

25 concepts · 259 links

optical dots ×3 · frozen silence ×3 · rigid profile ×3 · bourgeois leisure ×3 · social tension ×3 · zinc yellow ×2 · staged calm ×2 · class divide ×2 · no eye contact ×2 · painted border ×2 · parasol shadow ×2 · pet monkey ×2 · Egyptian frieze ×2

Same answers, six visual forms.



Lines	parallel axes
Isolines	poster contours
Spectrum	harmonic bars
Weave	a textile
Dots	Seurat field
Manifold	radar shape

Radar Manifold. Everyone’s four ratings layered on top of each other. Solid white is the Round 2 class average, dashed white is Round 1, orange is the pixel formula.

Play Save PNG 24 people · round 2 where given

Four ways to make work with data.

Print

Guerrilla Girls, *Do Women Have to Be Naked to Get into the Met. Museum?*
poster, 1989

W. E. B. Du Bois and students, *The Georgia Negro*
hand-drawn charts, Paris, 1900

Screen

Refik Anadol, *Unsupervised*
MoMA lobby, 2022–23

Aaron Koblin, *Flight Patterns*
animated flight paths, 2005

Object

Hans Haacke, *MoMA Poll*
two transparent ballot boxes, 1970

Nathalie Miebach
sculptures built from weather data

Drawing

Giorgia Lupi and Stefanie Posavec, *Dear Data*
postcards, 2014–15

Mark Lombardi, *Narrative Structures*
graphite network drawings, 1990s

Starter kit.

Collect

MoMA collection `GitHub CSV`

Met Open Access `CSV + API`

Art Institute of Chicago `API`

Wikidata `query service`

Google Forms `your own survey`

Clean and draw

Google Sheets, OpenRefine `cleaning`

Datawrapper, RAWGraphs, Flourish `no code`

Observable Plot `a little code`

p5.js, D3.js `creative code`

paper and pencil `always`

Read

Amanda Wasielewski, *Computational Formalism* `MIT Press, 2023`

Cleveland & McGill, “Graphical Perception” `JASA, 1984`

Johanna Drucker, “Humanities Approaches to Graphical Display” `DHQ, 2011`

Numbers count what repeats. You still decide what it means.